

Unit 31: Natural Logarithms — Full Solutions

Warm-up

1. $\ln(8)$

$8 = 2^3$, so:

$$\ln(8) = \ln(2^3) = 3\ln 2$$

2. $\frac{d}{dx}[\ln(5x)]$

Chain rule: $u = 5x$, $u' = 5$.

$$\frac{d}{dx}[\ln(5x)] = \frac{5}{5x} = \frac{1}{x}$$

(Alternatively: $\ln(5x) = \ln 5 + \ln x$, so its derivative is $0 + \frac{1}{x} = \frac{1}{x}$ — same answer.)

3. $\frac{d}{dx}[\ln(x^2 + 1)]$

Chain rule: $u = x^2 + 1$, $u' = 2x$.

$$\frac{d}{dx}[\ln(x^2 + 1)] = \frac{2x}{x^2 + 1}$$

4. Solve $\ln(x) = 3$.

$$x = e^3$$

5. $\int \frac{3}{x} dx$

$$\int \frac{3}{x} dx = 3\ln|x| + C$$

6. $\ln(x^3 y^2)$

$$\ln(x^3 y^2) = \ln(x^3) + \ln(y^2) = 3\ln x + 2\ln y$$

Standard

7. Solve $\ln(x - 8) = 5$.

Exponentiate both sides:

$$x - 8 = e^5$$

$$x = e^5 + 8$$

$$8. \frac{d}{dx}[x \ln x]$$

Product rule: $f = x$ ($f' = 1$), $g = \ln x$ ($g' = \frac{1}{x}$).

$$\frac{d}{dx}[x \ln x] = 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1$$

$$9. \frac{d}{dx} \left[\frac{\ln x}{x} \right]$$

Quotient rule: $f = \ln x$ ($f' = \frac{1}{x}$), $g = x$ ($g' = 1$).

$$\frac{d}{dx} \left[\frac{\ln x}{x} \right] = \frac{\frac{1}{x} \cdot x - \ln x \cdot 1}{x^2} = \frac{1 - \ln x}{x^2}$$

$$10. \ln(x^2 \sqrt{x+3})$$

Simplify first:

$$\ln(x^2 \sqrt{x+3}) = \ln(x^2) + \ln((x+3)^{1/2}) = 2 \ln x + \frac{1}{2} \ln(x+3)$$

Differentiate:

$$\frac{d}{dx} = \frac{2}{x} + \frac{1}{2} \cdot \frac{1}{x+3} = \frac{2}{x} + \frac{1}{2(x+3)}$$

$$11. \int \frac{2x}{x^2+1} dx$$

Let $u = x^2 + 1$, $du = 2x dx$.

$$\int \frac{1}{u} du = \ln|u| + C = \ln(x^2 + 1) + C$$

(No absolute value needed here since $x^2 + 1$ is always positive.)

$$12. \text{ Solve } \ln(2x+1) = 0.$$

$$2x+1 = e^0 = 1$$

$$2x = 0 \quad \Rightarrow \quad x = 0$$

$$13. \frac{d}{dx}[\ln(\sin x)]$$

Chain rule: $u = \sin x$, $u' = \cos x$.

$$\frac{d}{dx} [\ln(\sin x)] = \frac{\cos x}{\sin x} = \cot x$$

Challenge

14. $y = \frac{x^5}{4} \ln x - \frac{x^4}{4}$

Differentiate the first term using the product rule: $f = \frac{x^5}{4}$ ($f' = \frac{5x^4}{4}$), $g = \ln x$ ($g' = \frac{1}{x}$).

$$\frac{d}{dx} \left[\frac{x^5}{4} \ln x \right] = \frac{5x^4}{4} \ln x + \frac{x^5}{4} \cdot \frac{1}{x} = \frac{5x^4}{4} \ln x + \frac{x^4}{4}$$

Differentiate the second term: $\frac{d}{dx} \left[-\frac{x^4}{4} \right] = -x^3$.

Combine:

$$\frac{dy}{dx} = \frac{5x^4}{4} \ln x + \frac{x^4}{4} - x^3$$

15. $y = \ln \left(\frac{1}{x^2 \sqrt{x+1}} \right)$

Simplify first:

$$y = \ln(1) - \ln(x^2 \sqrt{x+1}) = 0 - \left[2 \ln x + \frac{1}{2} \ln(x+1) \right] = -2 \ln x - \frac{1}{2} \ln(x+1)$$

Differentiate:

$$\frac{dy}{dx} = -\frac{2}{x} - \frac{1}{2} \cdot \frac{1}{x+1} = -\frac{2}{x} - \frac{1}{2(x+1)}$$

16. $y = \ln(3x^3 e^{x^2})$

Simplify first:

$$y = \ln 3 + \ln(x^3) + \ln(e^{x^2}) = \ln 3 + 3 \ln x + x^2$$

(using $\ln(e^{x^2}) = x^2$ directly, since $\ln$ and $e^{(\cdot)}$ undo each other).

Differentiate ($\ln 3$ is just a constant, so its derivative is 0):

$$\frac{dy}{dx} = \frac{3}{x} + 2x$$

17. Solve $2 \ln(x) + \ln(4) = \ln(36)$.

Combine the left side using log properties: $2\ln x = \ln(x^2)$, so:

$$\ln(x^2) + \ln(4) = \ln(36)$$

$$\ln(4x^2) = \ln(36)$$

Since the logs are equal, their insides must be equal:

$$4x^2 = 36 \Rightarrow x^2 = 9 \Rightarrow x = 3$$

(We take only the positive root, since x must be positive for $\ln x$ to be defined in the original equation.)

18. $\int \frac{x^2 + 1}{x} dx$

Split the fraction first:

$$\frac{x^2 + 1}{x} = \frac{x^2}{x} + \frac{1}{x} = x + \frac{1}{x}$$

Now integrate term by term:

$$\int \left(x + \frac{1}{x} \right) dx = \frac{x^2}{2} + \ln|x| + C$$

Applied

19. $P(t) = 500e^{0.05t} = 1000$

$$e^{0.05t} = \frac{1000}{500} = 2$$

$$0.05t = \ln 2$$

$$t = \frac{\ln 2}{0.05} = 20 \ln 2 \approx 13.86 \text{ years}$$

20. $A(t) = 100e^{-0.2t} = 25$

$$e^{-0.2t} = \frac{25}{100} = 0.25$$

$$-0.2t = \ln(0.25) = -\ln 4$$

$$t = \frac{\ln 4}{0.2} = 5 \ln 4 = 10 \ln 2 \approx 6.93 \text{ years}$$

21. $MC(x) = \frac{10}{x}$, from $x = 1$ to $x = 5$.

$$\int_1^5 \frac{10}{x} dx = 10 \left[\ln x \right]_1^5 = 10(\ln 5 - \ln 1) = 10 \ln 5 \quad (\text{since } \ln 1 = 0)$$

$$\approx 10(1.6094) \approx 16.09$$

Answer: total cost increase $\approx$ \$16.09 (exactly $10 \ln 5$).

22. $\frac{dV}{dt} = \frac{6}{t}$, from $t = 1$ to $t = e$.

$$\int_1^e \frac{6}{t} dt = 6 \left[\ln t \right]_1^e = 6(\ln e - \ln 1) = 6(1 - 0) = 6$$

Answer: total volume change = 6 gallons.